INCH-POUND
MIL-I-24092/12
4 May 1989

MILITARY SPECIFICATION SHEET

INSULATING RESIN, SOLVENTLESS, TWO COMPONENT, REACTIVE, FOR SPOT PATCHING AND EMERGENCY REPAIRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24092.

REQUIREMENTS:

1. Grade - PK: Patching kit, two components.

2. Class - 130 and 155. The cured resin shall meet the requirements of the specified thermal class when determined in accordance with MIL-I-24092.

3. Composition - IV. Composition IV consists of resins that are in solution with either an organic monomer or an organic diluent. Such materials are not normally considered solvents since they are reactive and become part of the cured polymer.

4. The varnishes shall meet the qualification and quality conformance requirements specified in tables I and II.

AMSC N/A FSC 5970

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TABLE I. Qualification requirements, grade PK.

Characteristics (tests)	Units	Requirements
Storage life, separate components	Months, minimum	18
Flash point	°C, minimum	90
Curing time at 23°C	Hours, maximum	3
Working life after mixing	Minutes, minimum	90
Build, as applied by brush	Mils, minimum	2.0 to 5.0
Dielectric strength: 96/23/50	Volts per mil, minimum	1000
96/23/96	Percent decrease from 96/23/50 value, maximum	35
24/23/water	Percent decrease from 96/23/50 value, maximum	45
Dissipation factor: at 23°C	Percent, maximum	0.10
Dielectric constant: at 23°C	Maximum	4.5
Hardness, cured for 24 hours at 23°C, plus additional heat	Shore D, minimum Shore D, minimum	60 75
Bond strength at 23°C	Pounds force, minimum	4
Weight loss on cure at 150°C	Percent, maximum	3
Hydrolytic stability	Percent decrease in bond strength, maximum	65

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TABLE I. Qualification requirements, grade PK - Continued.

Characteristics (tests)	Units	Requirements
Chemical resistance:		
Hydraulic fluid	Percent decrease in bond strength, maximum	50
Lubrication oil		50
Cleaning fluid		50
Distilled water		50
Detergent solution		50

TABLE II. Quality conformance requirements, grade PK.

Characteristics (tests)	Units	Requirements
Curing time	Hours, maximum	3.5
Working life after mixing	Minutes, maximum	80
Build, as applied	Mils, minimum	0.9
Hardness, 24 hours at 23°C,	Shore D, minimum	60
Bond strength at 23°C	Pounds force, minimum	3
Thermal endurance by weight loss at 150°C	Percent, maximum	3

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NOTE:

Part or Identifying Number (PIN). Part number shall be M24092/12-(dash number indicating thermal class). For example, M24092/12-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance.

Custodians:
Army - ER
Navy - SH
Air Force - 20

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
Army - ME
Navy - MC, AS

Preparing activity:
Navy - SH
(Project 5970-0661-12)

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